

Assignment 3 - Mini-Project

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Problem Statement

Electricity regulation in India is a state subject. Renewable energy policies — net metering capacity limits, open access thresholds, wheeling and banking charges, and subsidy schemes — are set independently by each State Electricity Regulatory Commission (SERC) and revised through frequent amendment orders. A business development or regulatory analyst evaluating a solar or battery storage project (e.g. "What is the net metering limit for a 2 MW commercial plant in Gujarat, and is banking allowed?") must manually search dozens of state-specific PDF orders and cross-check them against the latest amendments. This is slow, error-prone, and a wrong or outdated figure can materially affect a project's financial viability.

This is a good topic for a RAG project because the answers are domain-specific, state-specific, time-sensitive, and must be traceable to an exact clause for the answer to be trustworthy — exactly the conditions under which retrieval-augmented generation outperforms a general-purpose LLM, which would otherwise produce a fluent but unverifiable or outdated answer.

Proposed Approach

- **Ingestion:** Parse state solar/RE policy orders, SERC net-metering regulations, and MNRE central guidelines (5 states). Chunk by clause/section rather than fixed length; tag each chunk with metadata (state, issuing authority, document type, date).
- **Indexing:** Embed chunks into a vector store (ChromaDB/FAISS).
- **Query understanding:** Detect the target state(s) from the query and decompose multi-state comparison questions into sub-queries.
- **Retrieval:** Hybrid search (BM25 + dense embeddings) with metadata filtering to prevent cross-state contamination, followed by cross-encoder reranking.
- **Amendment handling:** Explicit check for later amendment orders covering the same clause, with temporal ranking so superseded provisions are flagged rather than silently returned.
- **Generation:** Answer strictly from retrieved context with clause number, order name and date cited; abstain if the context is insufficient.
- **Evaluation:** RAGAS metrics (faithfulness, context precision, answer relevance) on a hand-written test set of ~30 Q&A pairs with known-correct answers.
- **(Optional, agentic) LangGraph workflow:** classify query → route to state-specific retriever → check for amendments → verify answer is grounded → re-retrieve with a reformulated query if ungrounded → format with citations.

Tech Stack

- **Orchestration:** LangChain (LangGraph if the agentic extension is implemented), as covered in class.
- **Vector store:** ChromaDB or FAISS.
- **Retrieval:** BM25 (keyword) + dense embedding retriever, combined via hybrid search; cross-encoder reranker.
- **Embeddings / LLM:** Open embedding model for indexing; class-taught LLM for generation.
- **Evaluation:** RAGAS.
- **Interface (demo):** Streamlit.